LAB 5: Data Visualization and Analysis using Pandas & Seaborn

Procedure

Step 1:

Library Setup:
Pandas, NumPy, Matplotlib, and Seaborn are imported.

Visualization Style:
Seaborn theme 'whitegrid' and matplotlib style 'ggplot' are applied.

Confirmation:
The message 'Libraries imported successfully.' is printed.

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Setting the visual theme for all plots
sns.set_theme(style="whitegrid")
plt.style.use('ggplot')

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

Figure 5.1

Step 2:

Correlation Analysis:
A heatmap displays correlation between Cases, Recovered, Deaths, and Tests.

Observation:
Strong correlation between Cases, Recovered, and Deaths.

```python
plt.figure(figsize=(10, 8))
correlation = df[['Cases', 'Recovered', 'Deaths', 'Tests']].corr()
sns.heatmap(correlation, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Correlation Matrix of COVID-19 Metrics', fontsize=15)
plt.show()
```

<Figure size 1000x800 with 2 Axes>

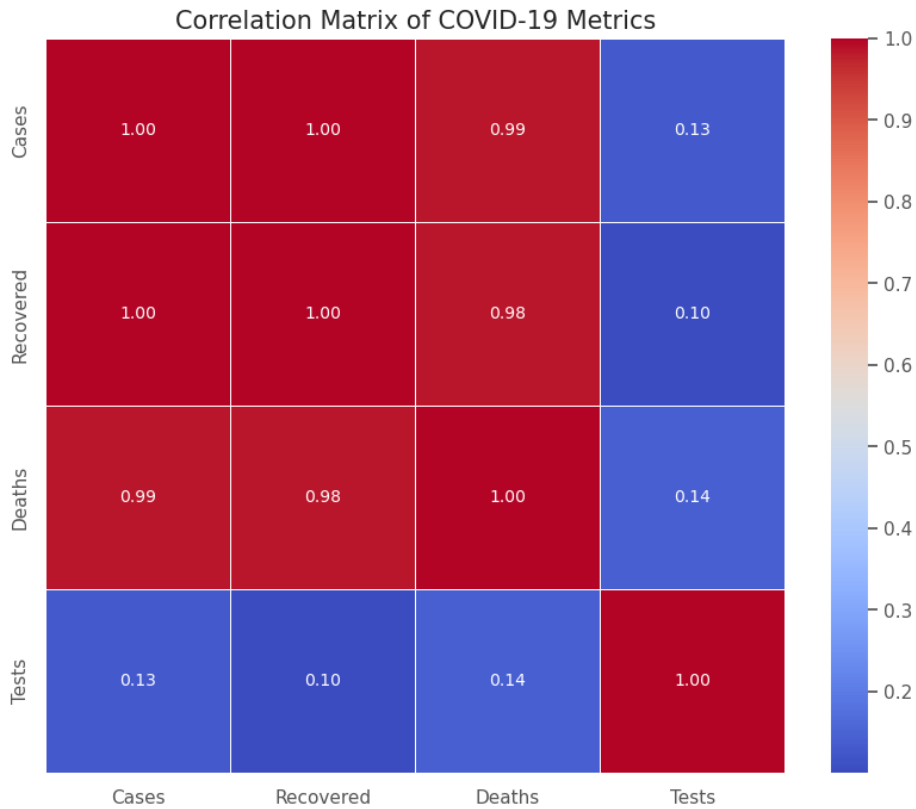

Figure 5.2

Step 3:

Regression Analysis:
Regression plots show relationships between Cases vs Recovered and Cases vs Deaths.

Insight:
Positive linear trends are observed.

```python
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Plot 1: Cases vs Recovered
sns.regplot(ax=axes[0], data=df, x='Cases', y='Recovered', color='blue')
axes[0].set_title('Regression: Cases vs Recovered')

# Plot 2: Cases vs Deaths
sns.regplot(ax=axes[1], data=df, x='Cases', y='Deaths', color='red')
axes[1].set_title('Regression: Cases vs Deaths')

plt.tight_layout()
plt.show()
```

<Figure size 1600x600 with 2 Axes>

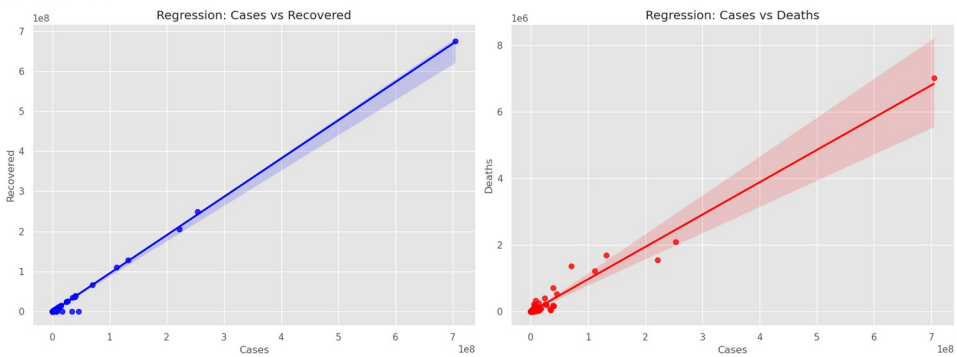

Figure 5.3

Step 4:

Pie Chart:

A pie chart shows proportion of COVID-19 cases by continent.

Observation:

Asia and Europe have the largest shares.

```python
continent_cases = df.groupby('continent')['Cases'].sum()
plt.figure(figsize=(8, 8))
plt.pie(continent_cases, labels=continent_cases.index, autopct='%1.1f%%', startangle=140, colors=sns.color_palette('pastel'))
plt.title('Proportion of COVID-19 Cases by Continent')
plt.show()
```

<Figure size 800x800 with 1 Axes>

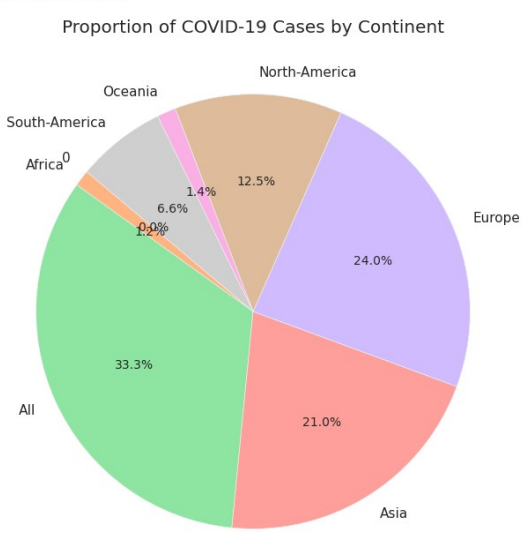

Figure 5.4

Step 5:

Time Series Aggregation:
A pivot table groups cases by continent over time.

Visualization:
Line plot shows cumulative cases progression.

```
df_pivot = df.pivot_table(index='day', columns='continent', values='Cases', aggfunc='sum')
df_pivot.plot.area(figsize=(12, 6), alpha=0.4)
plt.title('Cumulative COVID-19 Cases by Continent Over Time')
plt.ylabel('Cases')
plt.show()
```

```
<Figure size 1200x600 with 1 Axes>
```

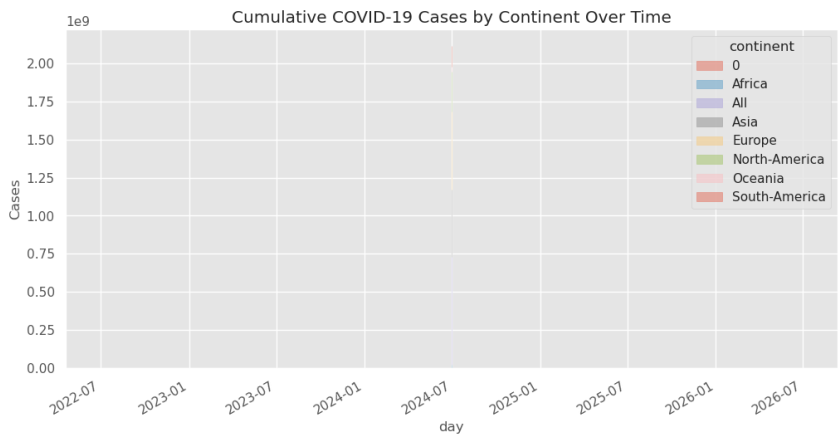

Figure 5.5

Step 6:

Country Count Analysis:
A bar chart shows number of countries represented per continent.
Insight:
Africa and Asia have higher counts compared to others.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='continent', data=df.drop_duplicates('country'), palette='viridis')
plt.title('Number of Countries Represented per Continent')
plt.show()
```

```
/tmp/ipykernel_1219323/443721073.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(x='continent', data=df.drop_duplicates('country'), palette='viridis')

<Figure size 1000x600 with 1 Axes>
```

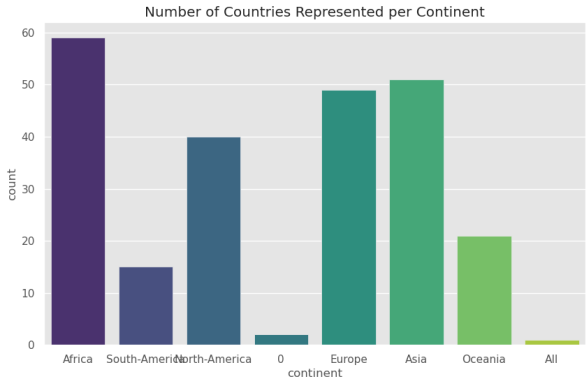

Figure 5.6

Step 7:

Dataset Loading:
The dataset 'data.csv' is loaded.

Preview:
The first 5 rows display columns such as country, continent, population, cases, recovered, deaths, and tests.

```python
# Load the dataset from CSV
df = pd.read_csv('data.csv')

# Display first 5 rows to verify the data
display(df.head())
```

```
       country   continent population      day \
0   Saint-Helena      Africa   6115.0 2024-06-30
1 Falkland-Islands South-America   3539.0 2024-06-30
2     Montserrat North-America   4965.0 2024-06-30
3 Diamond-Princess        NaN      NaN 2024-06-30
4   Vatican-City      Europe    799.0 2024-06-30

                 time Cases Recovered Deaths   Tests
0 2024-06-30T16:15:16+00:00  2166     2.0    NaN     NaN
1 2024-06-30T16:15:16+00:00  1930    1930.0   NaN  8632.0
2 2024-06-30T16:15:16+00:00  1403    1376.0   8.0 17762.0
3 2024-06-30T16:15:16+00:00   712     699.0  13.0     NaN
4 2024-06-30T16:15:16+00:00    29      29.0   NaN     NaN
```

Figure 5.7

Step 8:

Data Preprocessing:
The 'day' column is converted to datetime.

Cleaning:
Missing values are filled with 0.

Conversion:
Columns like Cases, Recovered, Deaths, Tests, and population are converted to numeric types.

Output:
Dataset info is printed showing structure and types.

```python
# Convert 'day' column to datetime
df['day'] = pd.to_datetime(df['day'])

# Handle missing values by filling with 0 (assuming no data means zero count in this context)
df = df.fillna(0)

# Ensure numeric columns are properly formatted
numeric_cols = ['Cases', 'Recovered', 'Deaths', 'Tests', 'population']
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric)

print("Data preprocessing complete.")
df.info()
```

```
Data preprocessing complete.
<class'pandas.core.frame.DataFrame'>
RangeIndex: 238 entries, 0 to 237
Data columns (total 9 columns):
 #  Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0  country    238 non-null    object
 1  continent  238 non-null    object
 2  population 238 non-null    float64
 3  day        238 non-null    datetime64[ns]
 4  time       238 non-null    object
 5  Cases      238 non-null    int64
 6  Recovered  238 non-null    float64
 7  Deaths     238 non-null    float64
 8  Tests      238 non-null    float64
dtypes: datetime64[ns](1), float64(4), int64(1), object(3)
memory usage: 16.9+ KB
```

Figure 5.8

Step 9:

Trend Visualization:
A line plot shows COVID-19 cases over time for selected countries.

Details:
Countries include Saint-Helena, Falkland-Islands, Montserrat, Diamond-Princess, and Vatican-City.

Output:
Trend lines with markers are displayed.

```python
plt.figure(figsize=(12, 6))
# Selecting a few countries for comparison
selected_countries = df['country'].unique()[:5]
df_filtered = df[df['country'].isin(selected_countries)]

sns.lineplot(data=df_filtered, x='day', y='Cases', hue='country', marker='o')
plt.title('COVID-19 Cases Trend for Selected Countries', fontsize=15)
plt.xlabel('Date')
plt.ylabel('Number of Cases')
plt.legend(title='Country')
plt.show()
```

```
<Figure size 1200x600 with 1 Axes>
```

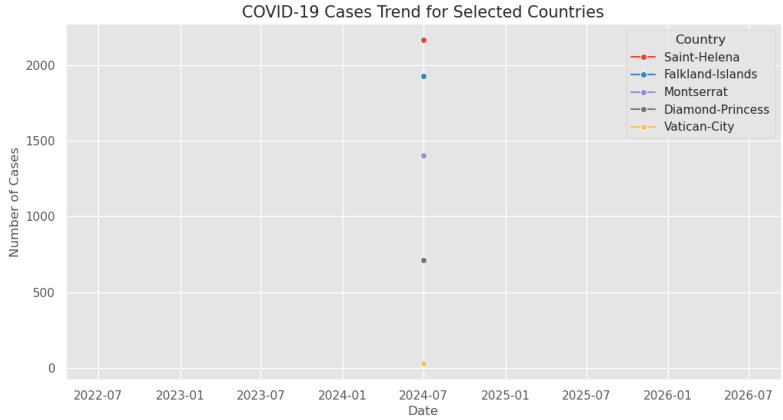

Figure 5.9

Step 10:

Top Countries Analysis:
Top 10 countries are selected based on total cases.

Visualization:
A horizontal bar chart displays countries and their case counts.

```
top_10_cases = df.sort_values(by='Cases', ascending=False).head(10)

plt.figure(figsize=(12, 6))
sns.barplot(data=top_10_cases, x='Cases', y='country', palette='viridis')
plt.title('Top 10 Countries by Total COVID-19 Cases', fontsize=15)
plt.xlabel('Total Cases')
plt.ylabel('Country')
plt.show()
```

```
/tmp/ipykernel_1219323/1529567383.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=top_10_cases, x='Cases', y='country', palette='viridis')

<Figure size 1200x600 with 1 Axes>
```

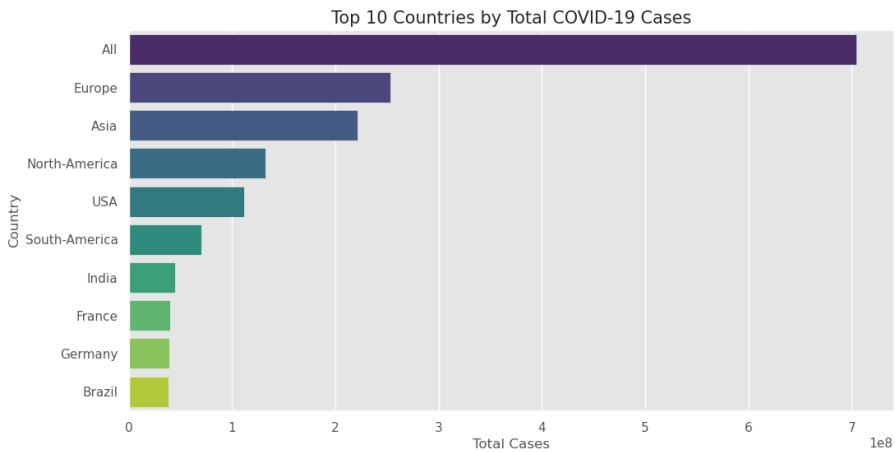

Figure 5.10

Step 11:

Distribution Analysis:
A histogram with KDE shows distribution of global COVID-19 cases.

Insight:
Most values are concentrated at lower ranges with a long tail.

```
plt.figure(figsize=(10, 6))
sns.histplot(df['Cases'], bins=20, kde=True, color='salmon')
plt.title('Distribution of COVID-19 Cases Globally', fontsize=15)
plt.xlabel('Cases')
plt.ylabel('Frequency')
plt.show()
```

```
<Figure size 1000x600 with 1 Axes>
```

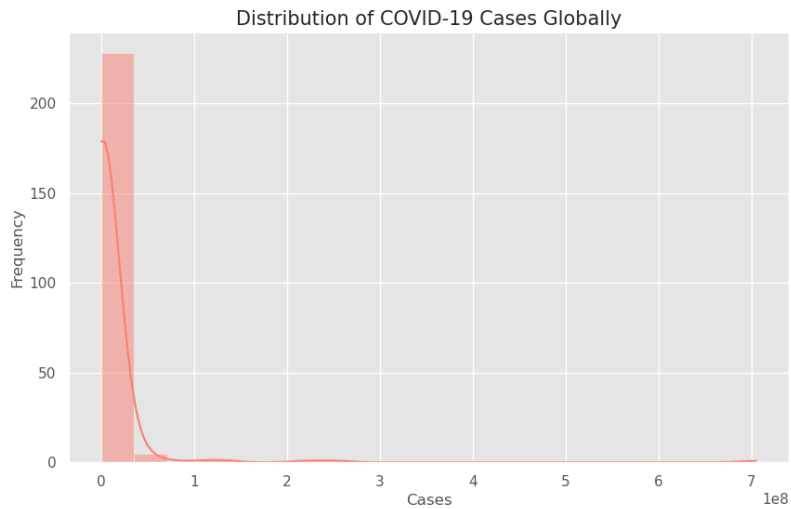

Figure 5.11

Step 12:

Scatter Analysis:
A scatter plot shows relationship between cases and deaths.
Details:
Points are colored by continent and sized by population.

```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Cases', y='Deaths', hue='continent', size='population', data=df, sizes=(50, 500))
plt.title('Relationship Between Cases and Deaths', fontsize=16)
plt.xlabel('Total Cases')
plt.ylabel('Total Deaths')
plt.show()
```

```
<Figure size 1000x600 with 1 Axes>
```

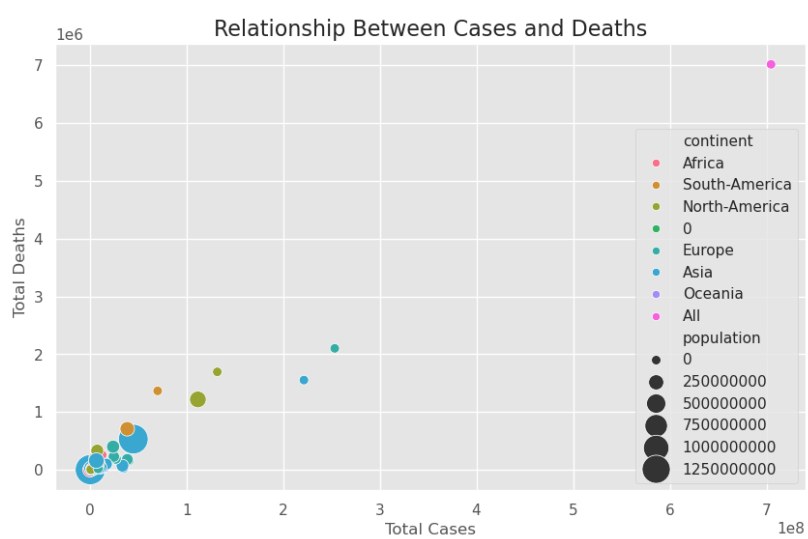

Figure 5.12

Step 13:

Box Plot:

Distribution of cases across continents is shown using a box plot.

Scale:

Log scale is used for better visualization.

```
plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x='continent', y='Cases', palette='Set2')
plt.title('Distribution of Cases by Continent', fontsize=15)
plt.xlabel('Continent')
plt.ylabel('Cases')
plt.yscale('log') # Log scale for better visibility
plt.show()
```

```
/tmp/ipykernel_1219323/3987660953.py:2: FutureWarning:

Passing  palette  without assigning  hue  is deprecated and will be removed in v0.14.0. Assign the  x  variable to  hue  and set  legend=False  for the same effect.

  sns.boxplot(data=df, x='continent', y='Cases', palette='Set2')

<Figure size 1200x600 with 1 Axes>
```

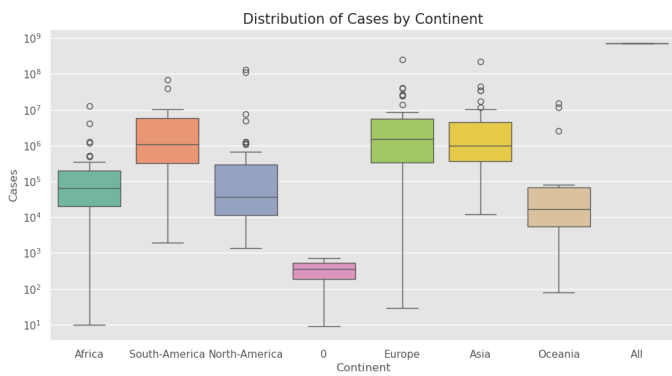

Figure 5.13

Step 14:

Violin Plot:

Distribution density of deaths across continents is visualized.

Insight:

Shows spread and concentration of death counts.

```
plt.figure(figsize=(12, 6))
sns.violinplot(data=df, x='continent', y='Deaths', palette='pastel')
plt.title('Distribution Density of Deaths by Continent', fontsize=15)
plt.xlabel('Continent')
plt.ylabel('Deaths')
plt.yscale('log')
plt.show()
```

```
/tmp/ipykernel_1219323/139793371.py:2: FutureWarning:

Passing  palette  without assigning  hue  is deprecated and will be removed in v0.14.0. Assign the  x  variable to  hue  and set  legend=False  for the same effect.

  sns.violinplot(data=df, x='continent', y='Deaths', palette='pastel')

<Figure size 1200x600 with 1 Axes>
```

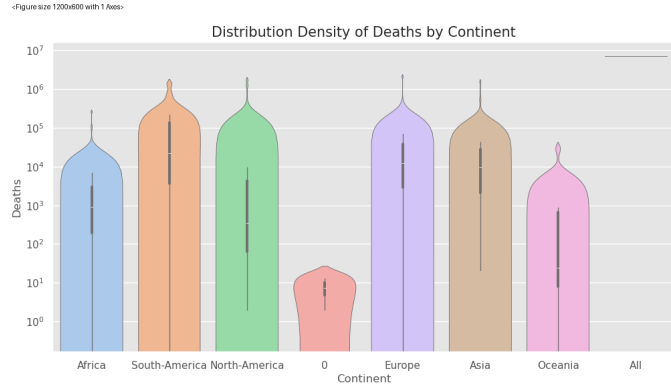

Figure 5.14